University of San Diego
SHILEY-MARCOS SCHOOL OF ENGINEERING

ADS-505 Project Start Form

Complete this form individually and submit to the Canvas assignment.

Your Idea

Business problem:
Local game stores (LGS) and secondary market vendors rely heavily on TCGplayer pricing benchmarks when purchasing card trade-ins from customers. Most stores calculate trade-in payouts as a flat percentage (e.g., 50% cash or 70% store credit) of the reported TCGplayer market price. However, the theoretical market price frequently diverges from the actual low floor price—the real-time price required to liquidate the card quickly. When cards suffer from wide price spreads (e.g., market is $20, but active low listings are $11), retailers overpay during trade-in intake and get trapped holding illiquid, declining stock. Counter staff lack an objective statistical tool to identify which card attributes correlate with high spread and liquidity risk to dynamically adjust buy-list offer margins.

Description of the data:
The project will use the open-source Pokemon-TCG-Data repository (curated by PokemonTCG.io via GitHub/API). The dataset spans over 15,000 card records across all historical English sets from 1999 Base Set through modern expansions.
Card Metadata: Card name, set series, set release date, artist, national Pokédex number, supertype (Pokémon, Trainer, Energy), and subtypes. Rarity & Finishes: Official rarity designation (Common, Rare Holo, Secret Rare, etc.) and finish variants (Normal, Holofoil, Reverse Holofoil). TCGplayer Market Metrics: Synchronized pricing tiers per finish including low, mid, high, and market prices in USD.

Your Project Goal

What is your motivation for choosing this project?
I enjoy trading card games. I have mostly spent time playing Magic: The Gathering, but Pokémon has a special place in my heart. The secondary market has priced out and turned off a lot of people. While supply-side decisions and artificial scarcity drive up prices, consumers cannot force publishers to print more cards and make games accessible (as much as I wish I could).

Instead, this project focuses on downstream challenges at the local game store counter. By analyzing real secondary market data, the goal is to make trade-in evaluations more accurate and efficient. This creates a fairer, more transparent experience for players selling their cards, helps local game stores avoid capital-draining dead stock, and keeps the local gaming community healthier overall.

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